Curriculum Vitae

Kwang Hyun Kim

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Web | **Google Scholar** | **ORCID**

Academic and Professional Background

2021 – Present | Ph.D. Candidate (Integrated Master's-PhD Program)

Ulsan National Institute of Science and Technology (UNIST), School of Energy and Chemical Engineering, Republic of Korea

- Advisor: Prof. Ja Hun Kwak (Former Senior Scientist at PNNL)
- Current Status: Passed dissertation proposal defense

2020 – 2021 (10 Months) | Intern Researcher, Korea Research Institute of Chemical Technology (KRICT), Daejeon, South Korea, Center for Environmental and Resource Research: Participated in research on catalyst development for LOHC (Liquid Organic Hydrogen Carriers) dehydrogenation to facilitate efficient hydrogen storage and transport.

2018 (Feb) – 2018 (June) | International Exchange Student, Vilnius Gediminas Technical University (VILNIUS TECH), Lithuania

2015 – 2017 | Military Service, Sergeant (Mandatory Fire Service), Republic of Korea, Mungyeong Fire Station, Gyeongsangbuk-do, South Korea

2014 – 2020 | B.S., Kyungpook National University (KNU), Daegu, South Korea, Bachelor of Engineering in Environmental Engineering

Graduate Research Experience

Graduate Research Assistant | UNIST (Advisor: Prof. Ja Hun Kwak) *2021 – Present*

Catalyst Development for CO/CO_2 hydrogenation

- **Action:** Joined a multi-year industrial project with a leading global chemical company and the steel industry, where I focused on the development of catalytic CO/CO_2 hydrogenation technology for high-value chemical manufacturing.
- **Impact:** Engineered a hybrid catalyst platform ($Na/ZnFe_2O_4$ and tailored zeolites) to break traditional scaling relations, achieving non-Schulz-Flory selectivity for value-added hydrocarbons.
- **Current Progress:** Investigated the synergistic effects of zeolite framework dimensionality and Al-siting, resulting in a manuscript recently submitted to Chemical Engineering Journal (2026).
- **Engineering Insight:** Validated catalyst durability and performance in fixed-bed reactors under simulated industrial conditions, providing critical data for process scale-up and reactor design.

Structural-defined Active Sites for Enhanced Stability (*Nature Communications*, 2025)

- **Action:** Developed modified TS-1 zeolites featuring structurally-defined dinuclear titanium sites (Ti-O-Ti) to mitigate framework degradation in extreme alkaline environments (pH 11).
- **Impact:** Demonstrated a 2.5-fold increase in catalytic activity and enabled 24-hour continuous operation, providing experimental benchmarks for structure-stability correlations in predictive catalyst design.
- **Method:** Utilized UV-Vis spectroscopy and XAS to investigate the active site evolution and structural integrity under harsh reaction conditions.

Bifunctional Catalyst Engineering for Tandem Processes (*Catalysis Today*, 2025)

- **Action:** Developed a **bifunctional Al-TS-1 catalytic system** by integrating Ti (epoxidation) and Al (hydrolysis) sites within a single MFI framework to shorten intermediate diffusion paths.
- **Impact:** Achieved **>95% Propylene Glycol (PG) selectivity** in a one-pot tandem process, effectively **addressing** the mass transfer limitations observed in traditional physical mixtures.

Publications

First-Authored Peer-Reviewed Journals

- **Kwang Hyun Kim**, et al., "Effects of Zeolite Topology and Acid Site Distribution on CO_2 Hydrogenation," ***Chemical Engineering Journal***, Submitted, (2026).
- **Kwang Hyun Kim**, SW Hwang, T Kim, H Kim, M Ko, SS Yoon, MS Kang, W Jin, et al., "Self-driven propylene epoxidation on modified titanium silicalite-1 by in situ generated hydrogen peroxide," ***Nature Communications***, 16 (1), 8636, (2025).
- **Kwang Hyun Kim**, Haneul Kim et al., "Direct production of propylene glycol from propylene using bifunctional Al-TS-1," ***Catalysis Today***, 464, (2025).
- EC Ra, **Kwang Hyun Kim**, JH Lee, S Jang, HE Kim, JH Lee, EH Kim, H Kim, et al., "Selective Light Hydrocarbon Production from CO_2 Hydrogenation over $Na/ZnFe_2O_4$ and CHA-Zeolite Hybrid Catalysts," ***ACS Catalysis***, 14 (5), 3492-3503, (2024).

Co-Authored Peer-Reviewed Journals

- Haneul Kim, **Kwang Hyun Kim**, Joonwoo Kim, Ja Hun Kwak, "Enhancing sulfur tolerance in Pt/TiO_2 catalysts: Effect of H_2O in CO oxidation," ***Chemical Engineering Journal***, 527, 171876, (2025).
- S Jang, D Gun Oh, H Kim, **Kwang Hyun Kim**, K Khivantsev, L Kovarik, et al., "Controlling the phase transformation of alumina for enhanced stability and catalytic properties," ***Angewandte Chemie***, 136 (15), e202400270, (2024).
- EC Ra, S Jang, DG Oh, JH Lee, HE Kim, EH Kim, MH Lee, **Kwang Hyun Kim**, et al., "A versatile hybrid catalyst platform of $Na/ZnFe_2O_4$ and zeolite for selective hydrocarbon production from CO_2 hydrogenation," ***Chemical Engineering Journal***, 470, 144335, (2023).

Awards and Honors

Outstanding Poster Award (Catalysis Division) | Apr 2025 | *2025 KIChE Spring Meeting, The Korean Institute of Chemical Engineers (KIChE)*

- Awarded for the excellence of the poster presentation in the field of catalysis.

Heart Saver Certification | Jun 2017 | Gyeongsangbuk-do Provincial Government, South Korea

- Awarded for life-saving emergency medical response and successful resuscitation of a cardiac arrest patient during mandatory fire service.

Presentations

Oral Presentations

- **Kwang Hyun Kim**, Haneul Kim, and Ja Hun Kwak, "**Self-driven Propylene Epoxidation Based on In-situ Hydrogen Peroxide Generation over Modified TS-1 Catalyst**," *2025 KIChE Fall Meeting, The Korean Institute of Chemical Engineers*, Jeju, South Korea (Oct 2025).

Poster Presentations

- **Kwang Hyun Kim**, Haneul Kim, and Ja Hun Kwak, "**Direct Production of Propylene Glycol From Propylene Using Bifunctional Al-TS-1**," *The 20th Korea-Japan Symposium on Catalysis*, Yonago, Japan (May 2025).
- **Kwang Hyun Kim**, Haneul Kim, and Ja Hun Kwak, "**Efficient Catalytic Conversion of Tri-Propylene Glycol to Di-Propylene Glycol Using Beta-Zeolite**," *2025 KIChE Spring Meeting, The Korean Institute of Chemical Engineers*, Daegu, South Korea (Apr 2025).

Patents

- **Kwang Hyun Kim,** Ja Hun Kwak, et al., "**A METHOD FOR HYDROLYSIS OF TRIPROPYLENE GLYCOL USING BETA-ZEOLITE**" *Korean Patent Application No. 10-2026-0068984, Filed on April 16, 2026.*